

Scenario:

Detective Angela Lansbury has handed you a case file with the below information and provided two logical images and two sets of registry files. The files are from a hotel lobby computer and a computer belonging to the subject.

The Monarch is a super villain whose costumed capers have attracted the attention of your police organization. The Monarch provided his busted laptop to Conjectural Technologies for repairs. Their technician, Pete White, alerted law enforcement to suspicious activity and acquired his laptop on behalf of officers with a warrant. Additionally, an image of a Clarion Hotel Internet kiosk has been provided on consent. It is believed the Monarch used this computer while on the run.

Your investigator, Agent Sampson, has asked you to provide information relating to a cold case on the criminal activity of supervillain Malcom Monarch. This may include communications, web searches and media related to his activity. A warrant covering all aspects of Agent Sampson's request has been signed by a North Carolina Superior Court judge.

Forensic Acquisition & Exam Preparation:

Having booted the forensic workstation to Windows 11, I verified the forensic files using FTK Imager 8.2.0 and Autopsy 4.23.1

Following this, all request items are satisfied using FTK Imager 8.2.0 and Autopsy 4.23.1

Findings and Report (Forensic Analysis)

Using FTK imager and Autopsy, I answered all the questions asked by the referring agency.

Each item is addressed separately below.

- 1) Was there communication from someone claiming to be the Guild's operative, or can you identify the operative's name/email?

To verify if there was communication from someone claiming to be the Guild's operative, or if the operative's name/email could be found was verified using Autopsy 4.23.1 with using the communications visualization tab. Using the communications visualization and selecting "MonarchIMG2021.E01 as the device selected, the following information was found. The emails were sent by MightyMonarch@outlook.com. There was one sent to the generalconsolidatedindustries@aol.com, this email was from someone claiming to be part of the Guild's operative; his name is Hamilton J Fantomos.

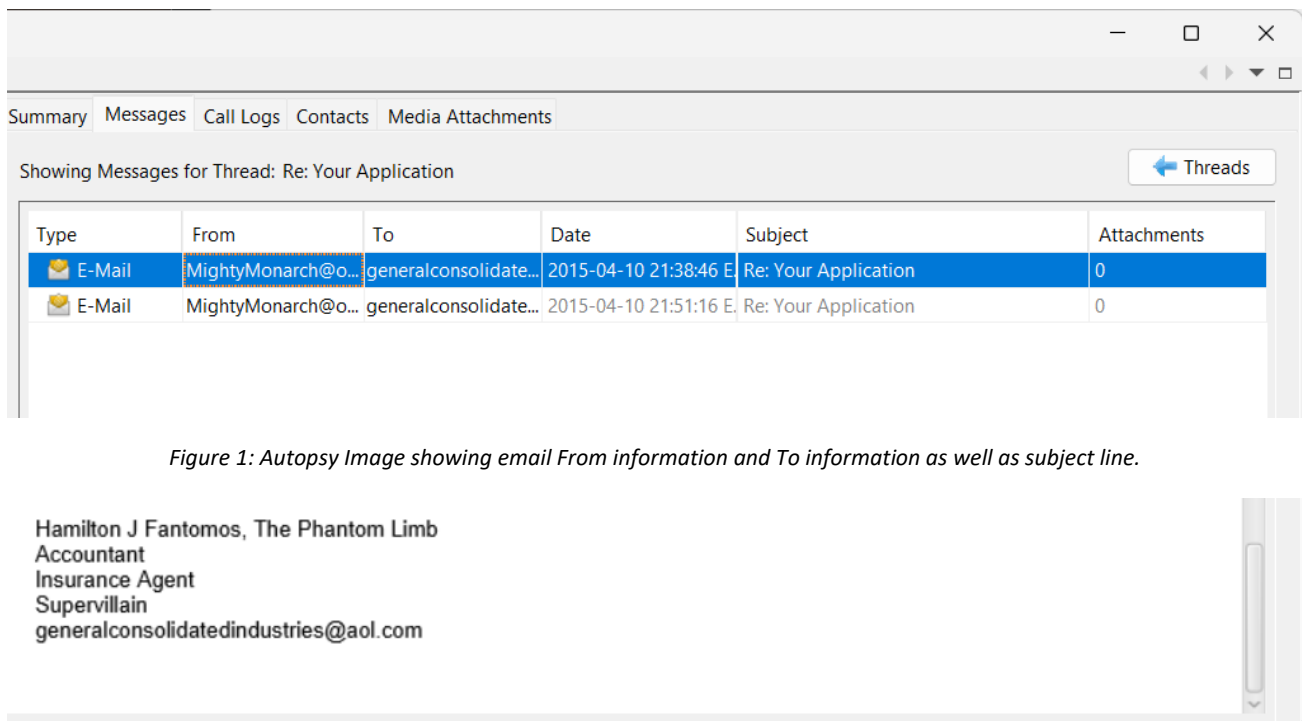


Figure 1: Autopsy Image showing email From information and To information as well as subject line.

Figure 2: Autopsy Image showing the name of the Guild's operative and email.

2) What is Malcolm's email address? Did he send an email to someone other than the guild operative? Who?

Using Autopsy, I was able to view Malcolm's email address, which is MightyMonarch@outlook.com, and Malcolm did email someone else named doctormrsthemonarch@outlook.com. Within this email, he stated, "Look who's on the New Yorker!! Menacing Venture only gets better now that it's Guild Of Calamitous Intent sanctioned!" While sending that message, he also sent the photo below.



Figure 3: Autopsy image showing the photo sent from Malcolm to "doctormrsthemonarch".

3) What version OS was running on the computer in use?

Using FTK Imager, viewing MonarchIMG.E01, going through to Monarch in this first partition, under Users followed by Malcom, the downloads folder under Desktop and Documents, the "images.jpg.FileSlack" within the hexadecimal text translated on the right-hand side, it states Windows 7 x86.

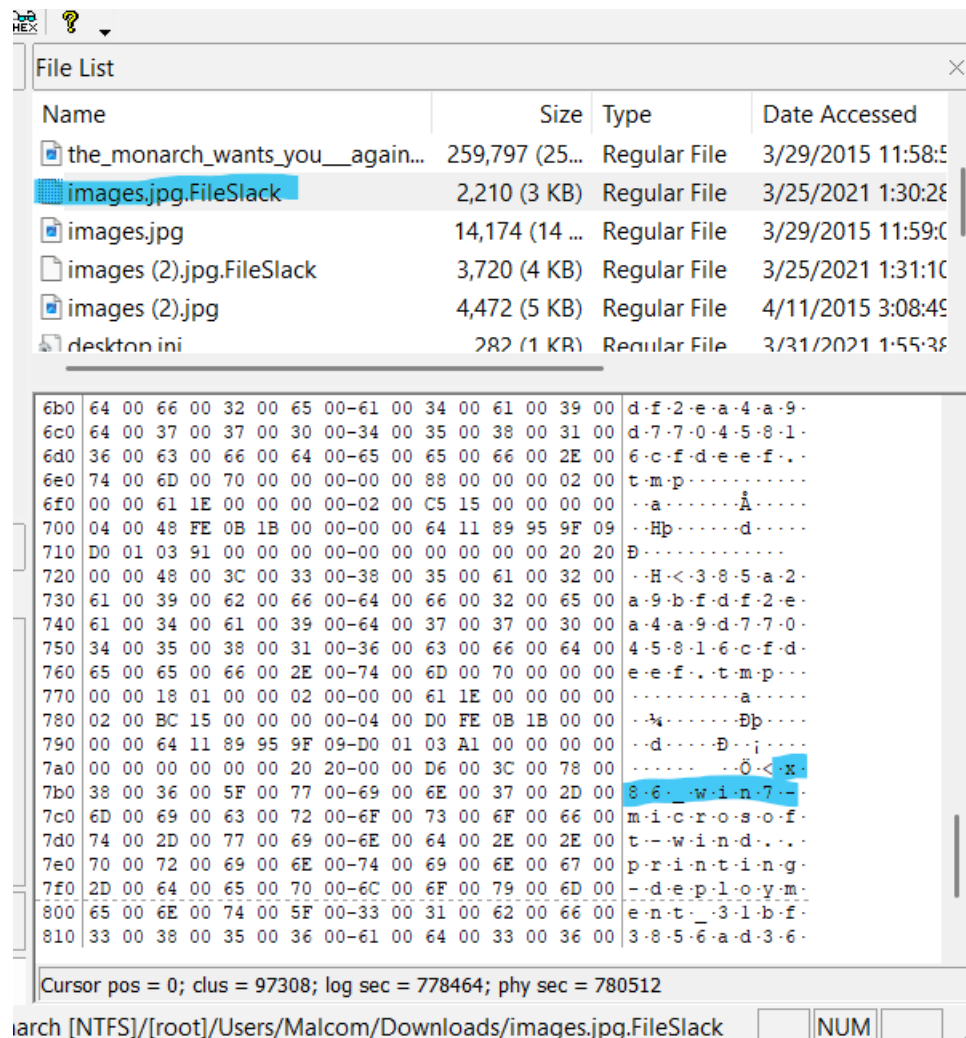


Figure 4: FTK Imager Hexadecimal translation of "images.jpg.FileSlack".

- 4) Did Malcon ever search for any terms that might mean he was going to cause any explosions? Using Autopsy within the bookmarks tab of the Monarch partition, the following websites were found. There are multiple instances in the bookmarks that show he was very interested in explosions, such as making them or buying them. The following websites bookmarked were "Big Fireworks Wholesale", "How to make Fireworks", "Bomb making instructions on the internet".

Big Fireworks® Wholesale and Retail Fireworks.url	1	https://www.bigfireworks.com/eSource3/default.aspx	Big Fireworks® Wholesale and Retail Fireworks.url	2015-03-29 20:26:44 EDT	Internet
How to Make Fireworks.url	1	http://www.instructables.com/id/How-to-Make-Firewo...	How to Make Fireworks.url	2015-03-29 20:22:41 EDT	Internet
Web Slice Gallery.url	1	http://go.microsoft.com/fwlink/?LinkId=121315	Web Slice Gallery.url	2013-10-23 12:23:09 EDT	Internet
My First Drone – Info On Building, Buying Or Flying	1	http://myfirstdrone.com/	My First Drone – Info On Building, Buying Or Flying Dro...	2015-03-29 20:18:02 EDT	Internet
Bomb-making instructions on the internet - Wikiped	1	http://en.wikipedia.org/wiki/Bomb-making_instructions.	Bomb-making instructions on the internet - Wikipedia, t...	2015-03-29 20:19:08 EDT	Internet
Suggested Sites.url	1	https://ieonline.microsoft.com/#ieslice	Suggested Sites.url	2013-10-23 15:30:01 EDT	Internet

Figure 5: Autopsy image of bookmarks tab of Monarch partition.

5) How about how to cover his tracks?

Through FTK Imager 8.2.0, through the Desktop folder under the Malcolm User, there is one folder named "New Folder"; through that folder is the "AllMyPlans.zip," which was encrypted. This is one of the only files encrypted, while others are just unallocated or can be clearly seen. Even though the files cannot be seen directly with what they contain, what can be seen is the types of files as well as their names. The following files are as follows: "bombs.gif", "Build Your Own Drone _ Flight Today _ Air & Space Magazine.pdf", two different "download.jpg", "drone_us_644x717.png", "drone1.jpg", "e0d128f8246b2bb256122c5030f0b05e.jpg", "German_Bombs_Shells.jpg", three different files named "images.jpg", "Lazy_dog_bombs.jpg", "milkweed toxin graph.gif", "MilkweedToxic SapThe Georgia Gardener.pdf", "toxic.jpg".

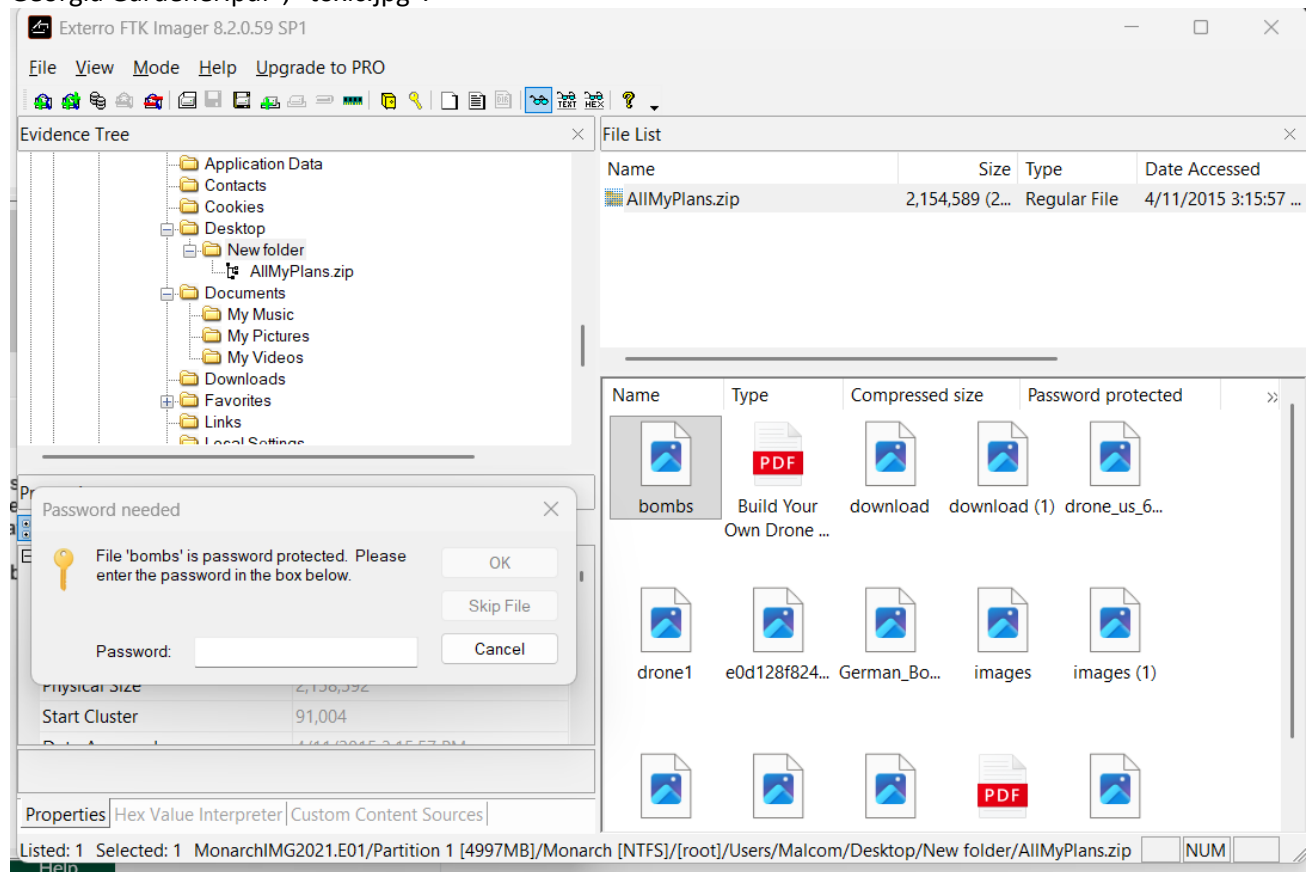


Figure 6: FTK Imager of Monarch/Users/Malcolm/Desktop/New Folder/AllMyPlans.zip.

6) Explain in detail what he was doing, with what software and when.

He used VirtualBox&VirtOS as a guest on

<https://www.modern.ie/vmhome?EIVersion=11&GuestOS=Win7&VirtPlatform=VirtualBox&VirtOS=Windows&VMBuild=20141027>.

- 7) Are there any photos that might have been downloaded? How do you know the photo was downloaded and not just moved to the folder where it was found? Is there a photo at this location that might tell us how he appears?

The image below, provided by Autopsy, shows a couple of files that were downloaded into the downloads root folder. This image proves that these photos were downloaded and not just moved into the folder because of the direct path. The next image shows, within Users/Malcolm/AppData/Local/Google/Chrome/User Data/Default/Cache/data_1, a possible photo of Malcolm. This possible photo of Malcolm was found using Timeline to view photos and other important information such as Web Cache.

History	1		https://www.google.com/url?sa=i&uact=j&q=&esrc=s&... google.com	Google Chrome	MonarchIMG2021.E01
History	1	C:\Users\JUser\Downloads\the_monarch_wants_you_...	http://files.chesscomfiles.com/images_users/tiny_mce/... chesscomfiles.com	Google Chrome	MonarchIMG2021.E01
History	1		https://www.google.com/url?sa=i&uact=j&q=&esrc=s&... google.com	Google Chrome	MonarchIMG2021.E01
History	1	C:\Users\JUser\Downloads\images.jpg	https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd... gstatic.com	Google Chrome	MonarchIMG2021.E01
History	1	C:\Users\JUser\Downloads\venture-bros-monarch.jpg	http://www.venturebrosblog.com/wp-content/uploads/. venturebrosblog.com	Google Chrome	MonarchIMG2021.E01

Figure 7: Autopsy image of the history tab showing which photos have been downloaded and which path they took.

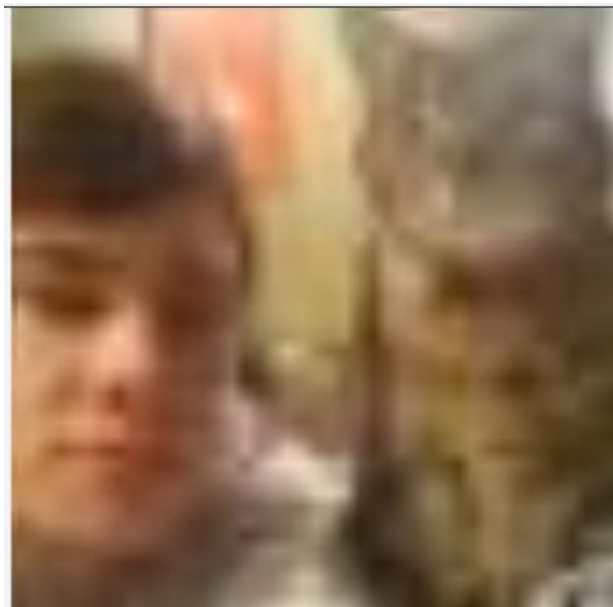


Figure 8: Autopsy recovered image of possible photo of Malcolm.

- 8) Did Malcolm do any research into forensics software? What specifically?

Using Autopsy and looking at Malcolm's search history, it could be seen that he looked up multiple ways "how to hide evidence" as well as "how to hide evidence computer". He also made a search for CCleaner, which is said to be a utility tool that cleans up "junk" files, fixes system errors, and deletes browsing history. He also had a download for Thunderbird, which is a known tool for emails and known for its "end-to-end" encryption techniques on emails. He used basic web searching techniques to look up how to hide evidence but never really went more in-depth with it.

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WebCacheV01.dat		bing.com	how to hide evidence	Microsoft Edge Analyzer	2015-03-29 23:19:12 EDT	MonarchIMG2021.E01
WebCacheV01.dat		bing.com	ccleaner	Microsoft Edge Analyzer	2015-03-30 00:01:46 EDT	MonarchIMG2021.E01
WebCacheV01.dat		bing.com	how to build a drone	Microsoft Edge Analyzer	2015-03-30 00:18:04 EDT	MonarchIMG2021.E01
WebCacheV01.dat		bing.com	organized villainy	Microsoft Edge Analyzer	2015-03-30 00:21:39 EDT	MonarchIMG2021.E01
WebCacheV01.dat		bing.com	how to make fireworks	Microsoft Edge Analyzer	2015-03-30 00:22:41 EDT	MonarchIMG2021.E01
WebCacheV01.dat		bing.com	how to hide evidence	Microsoft Edge Analyzer	2015-03-29 23:19:12 EDT	MonarchIMG2021.E01
WebCacheV01.dat		bing.com	how to hide evidence computer	Microsoft Edge Analyzer	2015-03-29 23:19:16 EDT	MonarchIMG2021.E01
WebCacheV01.dat		bing.com	ccleaner	Microsoft Edge Analyzer	2015-03-29 23:19:47 EDT	MonarchIMG2021.E01

Figure 8: Autopsy image of WebCache history showing multiple instances of "how to hide evidence", "ccleaner", and other suspicious searches

- 9) Is there any evidence he may have opened photos locally with an internet browser? Are those photos still available on the system?

Using Autopsy and the downloads history tab to view the Monarch partition shows that these photos had been opened locally and were downloaded onto the system. These photos are still available on the system.

History	1		https://www.google.com/url?sa=i&rct=j&q=&esrc=s&... google.com	Google Chrome
History	1	C:\Users\IEUser\Downloads\the_monarch_wants_you_...	http://files.chesscomfiles.com/images_users/tiny_mce/... chesscomfiles.com	Google Chrome
History	1		https://www.google.com/url?sa=i&rct=j&q=&esrc=s&... google.com	Google Chrome
History	1	C:\Users\IEUser\Downloads\images.jpg	https://encrypted-tbn0.gstatic.com/images?q=tbn:AND... gstatic.com	Google Chrome
History	1	C:\Users\IEUser\Downloads\venture-bros-monarch.jpg	http://www.venturebrosblog.com/wp-content/uploads/. venturebrosblog.com	Google Chrome

Figure 9: Autopsy image of downloads history from Monarch partition.



Figure 10: Autopsy image of "venture-bros-monarch.jpg" which is also in the downloads history.

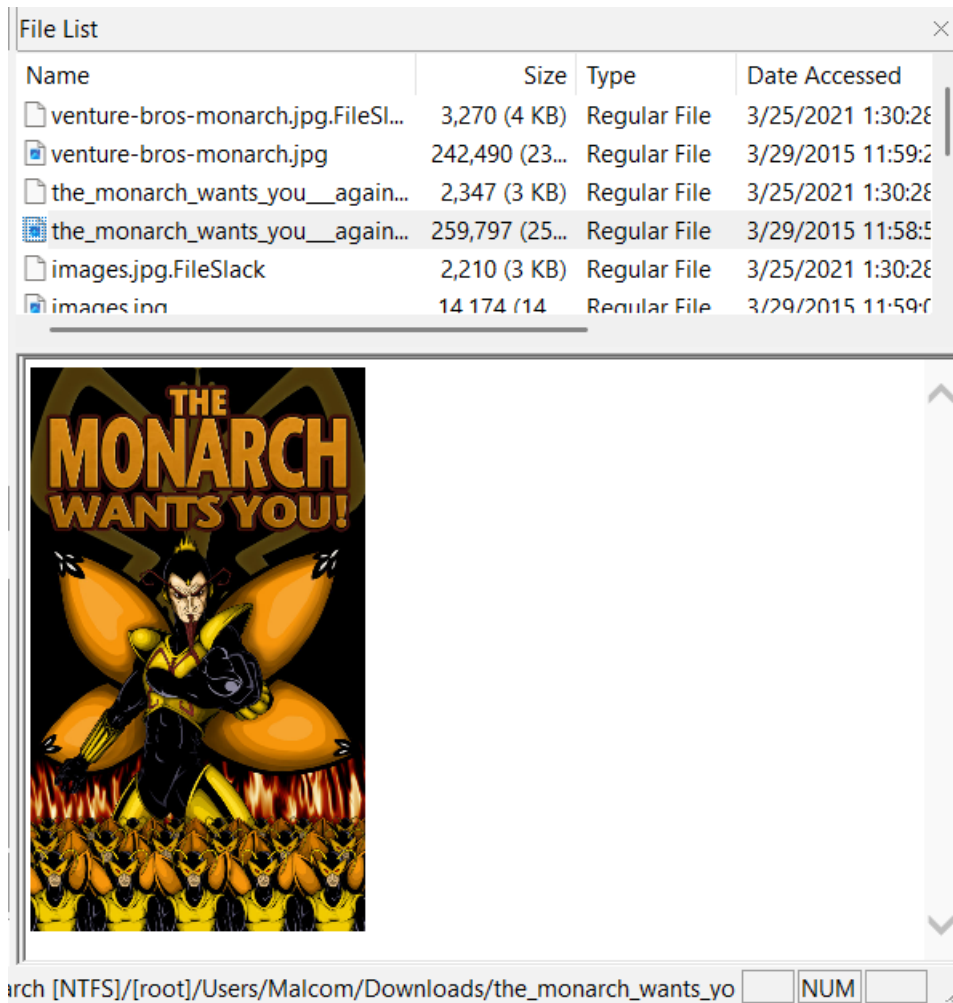


Figure 11: Autopsy image of "the_monarch_wants_you_again.jpg" which is also in the downloads history.

Additional Information

Some photos were uncovered using Autopsy within the EXIF metadata. Within those photos is one that looks like different-sized tubing, and another of what appears to be a homemade bomb. The images below are the names of the photos as well as the photos themselves.

Listing													
EXIF Metadata													
Table Thumbnail Summary													
Save Table as CSV													
Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Date Created	Device Model	Device Make	File Path	Size
FIXWGUF68BAFY.LARGE[1].jpg				File	0	Not Notable			2007-06-25 16:46:23 EDT	W800i	Sony Ericsson	/img_MonarchIMG2021.E01/vol_vol2/Users/Malcom/Ap	60818
FSRP4MTF68BAFLR.MEDIUM[1].jpg				File	0	Not Notable			2007-07-14 14:54:10 EDT	W800i	Sony Ericsson	/img_MonarchIMG2021.E01/vol_vol2/Users/Malcom/Ap	59426

Figure 12: Autopsy image of two different jpg images uncovered within the EXIF metadata.



Figure 13: Autopsy recovered image of different sized tubing.



Figure 14: Autopsy recovered image of what looks to be a homemade bomb.

The following emails were recovered from the Monarch partition sent to the `generalconsolidatedindustries@aol.com` in the image below. Monarch wrote many threats to the `generalconsolidatedindustries`, such as saying that he will “make your head match your arms if you know what I mean”, another threat was “I’m going to poison Venture, then blow him up with a robot drone, and after that I’m going to have plenty of free time to go work on little twerps like you!”. These were very serious threats that should not be taken lightly. Figure 16 consists of Monarch threatening Hamilton again, a Guild member, by stating “I’ve got plenty of bombs for my assault on Venture, and more than enough to blow all of you to kingdom come!”, another threat was “Prepare for murder! I’m going to slaughter you and all your filthy henchmen!”.

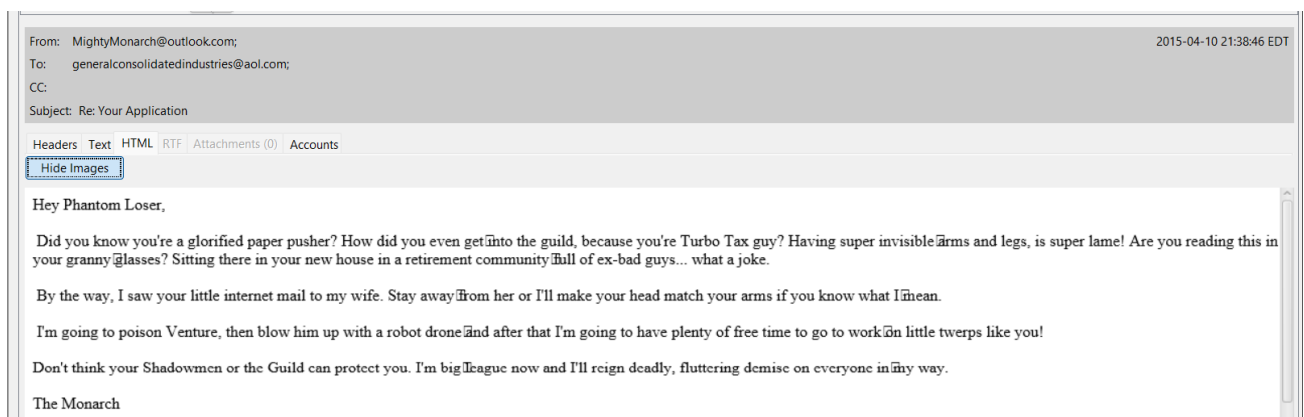


Figure 15: Autopsy image of threats sent from Monarch to Hamilton.

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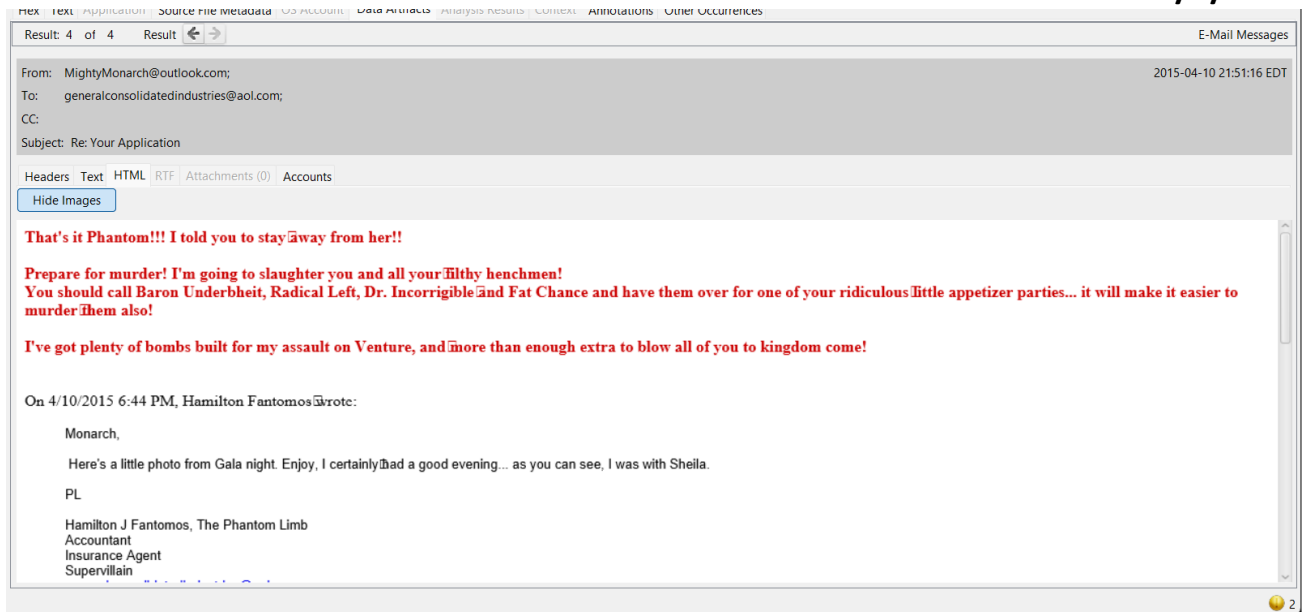


Figure 16: Autopsy image of threats sent from Monarch to Hamilton.

Conclusion

With this current analysis, it is possible that the Monarch had a large interest in harming others and creating explosions; there is also evidence where he threatened others, as well as evidence in his search history where he was looking for how to create bombs, as well as fireworks, or where to purchase them. In some emails, he states that he has created many bombs to either hurt or kill people.

Signed:

Sarah Lopez Martinez

Student Sarah Lopez Martinez

Student Forensics Examiner

Appendix

File System: The organization of files and folders allocated to the user as they appear on a hard drive, thumb drive or camera card. File systems retain information about files like time, path, name and other meta information.

Hash value: A mathematical formula known as a Hash Algorithm can be applied to any piece of data to prove it is identical to another piece of data. Altering data by even a single bit will radically change its hash value.

Byte: 1024 bits (a bit is 1 or 0). Files, file systems and physical disks are represented by their size in bytes, kilobytes (KB), megabytes (MB), gigabytes (GB), terabytes (TB). Each order of bytes is comprised of 1000 of the previous order of bytes. Some systems count by the true "byte" value for 1000 instead, using orders of 1024. This number's significance is due to binary math, where increasing values of 1 or 0 are called "base 2". Thus, in order to count to 1024, 2^{10} bits are required.

Forensic Image: A file containing every bit from an original physical disk that can be compared by hash to be mathematically identical to the original item. Changing this image by even 1 bit will cause the hash values to be different.

Zone.Identifier: Is a value assigned in a Alternate Data Stream to a file in NTFS on Windows Operating systems to determine the source of a file on various networks.

Alternate Data Stream: An additional amount of data that is ad hoc sized for various purposes by either the Operating System or software and is not visible to the user and cannot be copied to non-NTFS file systems.

FAT32: A type of file system.